FINITE FREE RESOLUTIONS WITHOUT BASES: THE STRUCTURE THEOREM OF BUCHSBAUM AND EISENBUD AND THE GCD

Abstract. Coquand and Quitté [4] gave elementary constructive proofs of results of Buchsbaum and Eisenbud [1, 2] on finite free resolutions: a resolution of an ideal $(a_1, \dots, a_N)$ yields a greatest common divisor of the a_j, computed as the Cayley determinant of the resolution. The statements are about matrices, and the Cayley determinant depends on the bases, up to a unit. We show what this ambiguity is: the canonical object is a line with an injective map $a_1 : \mathrm{Det}(F) \to R$, and choosing a basis of the line produces an ordinary greatest common divisor. The regular greatest common divisors of the a_j are exactly the values of a_1 at the bases of $\mathrm{Det}(F)$, and the pair $(\mathrm{Det}(F), a_1)$ depends only on the ideal, up to a unique isomorphism. To obtain this we restate the results for finite free modules, in univalent foundations, following the form of the first structure theorem that Buchsbaum and Eisenbud give for projective modules [2, Theorem 3.1 and the Remark after it]: the multipliers a_k are maps between lines defined without tensor products or duals, and the type of such structures on a complex is contractible. The same method gives Wiebe's theorem [5] without bases. All results are checked in the proof assistant Unit.

1. Introduction

Let R be a commutative ring and $0 \to F_n \xrightarrow{f_n} \cdots \xrightarrow{f_1} F_0$ an exact complex of finite free modules. Buchsbaum and Eisenbud [2, Theorem 3.1] show that each $\bigwedge^{r_k} f_k$ factors through a map a_k into $\bigwedge^{r_k} F_{k-1}$ from a module of rank one, and that these factorisations are linked from one step to the next. In their remark after Theorem 3.1 they explain that for projective modules, which need not be oriented, the identifications $\bigwedge^{r_{k+1}} F_k \cong \bigwedge^{r_k} F_k^*$ have to be replaced by natural isomorphisms involving lines $R_k = \bigwedge^{s_k} F_k \otimes R_{k+1}^*$; the theorem then holds with maps $a_k : R_k \to \bigwedge^{r_k} F_{k-1}$.

Coquand and Quitté [4] prove these results constructively, with the notion of true grade instead of depth and without prime ideals, in coordinates: $F_k = R^{p_k}$, f_k a matrix A_k. Their Lemma 8.4 and Corollary 8.5 write $\bigwedge^{r_k}(A_k) = \beta_k \beta_{k+1}^{*T}$ with β_k in $\bigwedge^{r_k}(F_{k-1})$; the last vector β_1 is the Cayley determinant, a greatest common divisor of the r_1-minors of A_1 (Theorem 8.7), and a finite free resolution of $(a_1, \dots, a_N)$ gives a greatest common divisor of the a_j (Theorem 8.9). Two features are tied to the coordinates: the dual β^* is the Hodge dual, so the factorisation carries a sign $(-1)^{pq}$ depending on conventions, and the Cayley determinant depends on the bases up to a unit. The book of Coquand and Lombardi [3] gives a complete constructive treatment of Northcott's theory [9], including these results.

This paper gives the coordinate-free form, and with it an explanation of the ambiguity "up to a unit". For a resolution F of a nonzero ideal $(a_1, \dots, a_N)$ the canonical object is not an element of R but a line $\mathrm{Det}(F)$ together with an injective linear map $a_1 : \mathrm{Det}(F) \to R$ whose image is the gcd ideal. A basis φ of the line gives the element $a_1(\varphi)$, and

$$\mathrm{Bases}(\mathrm{Det}F) \;\simeq\; \mathrm{RegGcd}(a_1, \dots, a_N), \qquad \varphi \mapsto a_1(\varphi)$$

(Theorem 10): the regular greatest common divisors are exactly the values of a_1 at the bases, and the unit relating two of them is the unit relating two bases of a line. Moreover the pair

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$(\mathrm{Det}(F), a_1)$ depends only on the ideal: for two resolutions F, G there is a unique isomorphism $e : \mathrm{Det}(F) \cong \mathrm{Det}(G)$ with $a_G e = a_F$ (Theorem 14).

The setting is univalent foundations [8]: a finite free module is a module that is *merely* isomorphic to some R^n, with no basis chosen. The main steps are the following.

(1) The lines of Buchsbaum and Eisenbud can be written with Hom alone: $R_{n+1} = R$ and $R_k = \mathrm{Hom}(R_{k+1}, \bigwedge^{s_k} F_k)$. The condition on the a_k is

$$\bigwedge^{r_k} f_k(x) = a_k\big(\rho \mapsto x \wedge a_{k+1}(\rho)\big),$$

which uses only the exterior product. Under the hypotheses of [4, Corollary 8.5], stated with ideals of minors defined without bases, the type of families (a_k) satisfying it is contractible (Theorem 5). In coordinates the condition is exactly the factorisation of [4, Lemma 8.4] with no sign (Proposition 6).

(2) $\mathrm{Det}(F) := R_1$ is a line and a_1 is canonical. For a resolution of a nonzero ideal $(a_1, \ldots, a_N)$ the last rank is 1, so $a_1 : \mathrm{Det}(F) \to R$, and a_1 is injective. The regular greatest common divisors of the a_j are exactly the elements $a_1(\varphi)$, φ a basis of the line $\mathrm{Det}(F)$, and they form an $R^\times$-torsor (Theorem 10). The Cayley determinant is a_1 at the basis element determined by the chosen bases.

(3) Two resolutions of the same nonzero ideal, of any lengths, have canonically isomorphic determinant lines, compatibly with a_1 (Theorem 14); an isomorphism of resolutions is an equality of resolutions (structure identity principle), and the induced isomorphism is the canonical one (Theorem 15). A resolution also decides whether the ideal is zero (Proposition 11).

(4) The same method gives Wiebe's theorem for linear forms on a finite free module, with the determinant of an endomorphism defined on the top exterior power (Theorem 16).

The method is uniform: uniqueness lets coordinate constructions descend to canonical objects, and univalence makes their invariance automatic. An object attached to finite free modules is defined by a property that is a proposition, or a type that is contractible; it is then an element by unique choice. To prove the property it suffices to prove it after choosing bases, and there the matrix arguments of [4] apply. No construction depends on a choice of bases, and invariance under isomorphisms is transport.

2. Setting

We work in univalent type theory [8]. A type is a *proposition* if any two of its elements are equal; $\|A\|$ is the propositional truncation, and "merely A" means $\|A\|$. A type is *contractible* if it has an element equal to every element. *Unique choice*: if P is a proposition, $\|P\| \to P$; in particular an element of a contractible type, or of a proposition that is merely inhabited, is obtained without choice. Rings are commutative with a set as carrier; we do not assume $0 \neq 1$.

Modules and structure identity. For modules A, B the identity type $A = B$ is equivalent to the type of linear bijections $A \cong B$; equivalently the type $\sum_B (A \cong B)$ is contractible. (The type $\sum_B (A = B)$ is contractible for every type; the content of the statement is that the same holds with $\cong$.) This is the structure identity principle, proved once for a notion of structure built from operations, actions and points with axioms that form a proposition [6], via the fundamental theorem of identity types [7, Theorem 11.2.2]; modules and commutative rings are instances, and the only algebraic input is that their axioms are propositions (zero and negation are unique given associativity and commutativity). So to prove a statement about all pairs $(B, \varphi : A \cong B)$ it suffices to prove it for (A, id) (induction on isomorphisms). The same

holds for families: $\sum_{G'} \prod_i (G_i \cong G'_i)$ is a product of contractible types, hence contractible. For commutative rings, $\sum_{S'}(S \cong S')$ is contractible as well.

A module M is *finite free of rank n* if merely $M \cong R^n$; if $1 \neq 0$ the rank is unique (an injective linear map $R^{n+1+j} \to R^n$ forces $1 = 0$, by McCoy's theorem). A *line* is a module merely isomorphic to R. For modules K, M, $\mathrm{Hom}(K, M)$ is a module; if K and M are lines, so is $\mathrm{Hom}(K, M)$. An element φ of a module K is a *basis* if $c \mapsto c\varphi$ is bijective, i.e. $c\varphi = c'\varphi$ implies $c = c'$ and every element is $c\varphi$ for some c. The bases of a line form an $R^\times$-torsor.

Exterior powers. For a finite free M, the type of exterior algebras on M (an algebra with a linear map $M \to E$, $v^2 = 0$, universal for such maps) is contractible: on R^n the exterior algebra is constructed explicitly, and contractibility transports along $M \cong R^n$. We write $\bigwedge M$ for its centre. An element of $\bigwedge M$ has degree k if its coordinates have degree k in every basis; these elements form $\bigwedge^k M$, and a linear map $f : M \to N$ induces $\bigwedge f$ and $\bigwedge^k f$. If M has rank n, $\bigwedge^n M$ is a line, and in a basis e the top coordinate is an isomorphism $\bigwedge^n M \cong R$.

Grade. For a family $(b_i)_{i \in I}$ of elements of R, *regular* means $xb_i = 0$ for all i implies $x = 0$; $\mathrm{Gr}(b) \geq 2$ means regular and every family (y_i) with $b_i y_j = b_j y_i$ is merely of the form $y_i = tb_i$. This is the case $k = 2$ of the grade of [4, Section 6]; the true grade $\mathrm{Gr}(b) \geq k$ of a finite family is defined with regular sequences after adding indeterminates.

3. Ideals of minors and ranks without bases

Definition 1. Let $f : M \to N$ be linear between finite free modules and $k \geq 0$. The family $D_k(f)$ consists of the elements $\lambda(f(v_1) \wedge \cdots \wedge f(v_k))$, for $v_1, \ldots, v_k \in M$ and $\lambda : \bigwedge N \to R$ linear.

Proposition 2. *For bases of M and N, with A the matrix of f, every k-minor of A is an element of $D_k(f)$, and every element of $D_k(f)$ is in the ideal generated by the k-minors. Hence "$D_k(f)$ is regular" and "$\mathrm{Gr}(D_k(f)) \geq 2$" hold if and only if they hold for the k-minors of A, and do not depend on the bases.*

Proof. A minor is the coefficient at a k-subset of rows of the wedge of k columns; take the v_t to be the basis vectors of these columns and λ the corresponding coordinate. Conversely the coordinates of $f(v_1) \wedge \cdots \wedge f(v_k)$ are in the ideal of k-minors, and a linear form on $\bigwedge N$ is a combination of coordinates. Regularity and $\mathrm{Gr} \geq 2$ pass to a family b from a family a when every a_i is in the ideal of b, in the sense used by the definition of $\mathrm{Gr} \geq 2$ (using unique choice to build the factor t). □

Definition 3. f is *stable of rank r* if $\bigwedge^{r+1} f = 0$ (every $f(v_1) \wedge \cdots \wedge f(v_{r+1})$ is 0) and $D_r(f)$ is regular.

Both conditions are propositions. If f is stable of ranks $r < r'$ then $\bigwedge^{r'} f = 0$, so $D_{r'}(f) = 0$ is regular and $1 = 0$. Thus the rank is unique when $1 \neq 0$. In a basis, stability of rank r is the notion of [4, Section 2] for the matrix.

4. The structure theorem

Let $0 \to F_n \xrightarrow{f_n} F_{n-1} \to \cdots \xrightarrow{f_1} F_0$ be a complex of finite free modules, $f_k f_{k+1} = 0$, with numbers $r_k \geq 0$ ($1 \leq k \leq n+1$, $r_{n+1} = 0$) such that F_k has rank $s_k = r_k + r_{k+1}$ for $1 \leq k \leq n$.

Definition 4. The lines are $R_{n+1} = R$ and $R_k = \mathrm{Hom}(R_{k+1}, \bigwedge^{s_k} F_k)$. A *structure* on the complex is a family of linear maps $a_k : R_k \to \bigwedge^{r_k} F_{k-1}$, $1 \leq k \leq n+1$, with $a_{n+1} = \mathrm{id}_R$ ($\bigwedge^0 F_n = R$), such that for $1 \leq k \leq n$

$$(*) \qquad \bigwedge^{r_k} f_k(x) = a_k\big(\rho \mapsto x \wedge a_{k+1}(\rho)\big) \qquad (x \in \textstyle\bigwedge^{r_k} F_k),$$

and a_k is regular for $2 \le k \le n+1$: $c\,a_k(\varphi) = 0$ for all φ implies $c = 0$.

Here $x \wedge a_{k+1}(\rho) \in \bigwedge^{s_k} F_k$, so $\rho \mapsto x \wedge a_{k+1}(\rho)$ is an element of R_k. Since R_{k+1} and $\bigwedge^{s_k} F_k$ are lines, R_k is a line. For a line L and a line K, $L \otimes K^* = \mathrm{Hom}(K, L)$ canonically, and through this identification Definition 4 is the form of [2, Theorem 3.1(a)] given in the Remark after it, for modules that are not oriented: the natural isomorphism used there becomes the exterior product. For n we get $R_n \cong \bigwedge^{s_n} F_n$ and $a_n = \bigwedge^{r_n} f_n$, as in [2].

Theorem 5. *Assume* $\mathrm{Gr}(D_{r_k}(f_k)) \ge 2$ *for* $2 \le k \le n$ *and* $D_{r_1}(f_1)$ *regular. Then the type of structures on the complex is contractible.*

Only these hypotheses are used: that F is a complex, the rank equations $s_k = r_k + r_{k+1}$, and the two grade conditions. Neither exactness nor stability of the f_k is assumed. This is all that [4, Lemma 8.4 and Corollary 8.5] use: the composite of consecutive maps is 0, the sizes are $p+q$ and $q+s$, and the grade conditions on the factors; the proportionality of [4, Lemma 8.2] needs no stability. Exactness enters only in Section 5, where [4, Theorem 6.8] produces the ranks, stability and the grade conditions from it. By Proposition 2 the grade conditions are statements about the f_k, and in coordinates they are the hypotheses of [4] for any bases.

Proof. Uniqueness. If a_{k+1} is regular, $(*)$ determines a_k. This is a proposition, so we may choose bases; let ρ_k be the induced basis of the line R_k and $\beta = a_{k+1}(\rho_{k+1})$, in coordinates. The element $w_x : \rho \mapsto x \wedge a_{k+1}(\rho)$ of R_k is $\mathrm{top}(x \wedge \beta)\,\rho_k$. If a_k and a'_k both satisfy $(*)$, then $\mathrm{top}(x \wedge \beta)\,(a_k(\rho_k) - a'_k(\rho_k)) = 0$ for all x. Taking for x the basis vectors e_I of $\bigwedge^{r_k} R^{s_k}$, the scalars $\mathrm{top}(e_I \wedge \beta)$ are, up to sign, the coordinates of β, which form a regular family because a_{k+1} is regular. So every coordinate of $a_k(\rho_k) - a'_k(\rho_k)$ is 0, and $a_k = a'_k$ since ρ_k is a basis. By descending induction from $a_{n+1} = \mathrm{id}$, a structure is unique. Being a structure is therefore a proposition, and by unique choice existence may be proved after choosing bases.

Existence. Choose bases of the F_k. They give bases of the lines R_k. By [4, Corollary 8.5] there are β_k with $\bigwedge^{r_k}(A_k) = \beta_k \hat{\beta}^T_{k+1}$, normalised as in Proposition 6 below. Define a_k by $a_k(\text{basis element}) = \beta_k$. By Proposition 6, $(*)$ holds, and a_k is regular since β_k is ($\mathrm{Gr}(\beta_k) \ge 2$ for $k \ge 2$ in [4]). □

The comparison with [4] is a computation with the canonical pairing. For $x \in \bigwedge^q R^N$ and $y \in \bigwedge^p R^N$, $p+q = N$, let $\mathrm{top}(x \wedge y)$ be the coefficient of $e_1 \wedge \cdots \wedge e_N$. Define $\hat{y} \in \bigwedge^q R^N$ by $\hat{y}_I = \epsilon(I, I^c)\, y_{I^c}$ for $|I| = q$, where $e_I \wedge e_{I^c} = \epsilon(I, I^c)\, e_1 \wedge \cdots \wedge e_N$. Then $\mathrm{top}(x \wedge y) = \langle x, \hat{y} \rangle$, the coordinatewise scalar product. For a Hodge dual y^* defined with $\epsilon(I^c, I)$ in place of $\epsilon(I, I^c)$ one has $\hat{y} = (-1)^{pq} y^*$; this is the sign that the factorisation of [4, Lemma 8.4] depends on.

Proposition 6 (dictionary)**.** *Let* $E \xrightarrow{A} R^N \xrightarrow{B} G$ *in coordinates,* $\beta \in \bigwedge^p R^N$*, and* $\gamma \in \bigwedge^q G$*,* $N = p+q$*. The condition* $\bigwedge^q(B)(x) = \mathrm{top}(x \wedge \beta)\,\gamma$ *for all* x *of degree* q *(that is,* $(*)$ *in coordinates) holds if and only if* $\bigwedge^q(B) = \gamma \hat{\beta}^T$*. So* [4, Lemma 8.4]*, every stage of* [4, Corollary 8.5]*, and the Cayley determinant are instances of* $(*)$*, with* $\hat{\beta}$ *in place of* β^**.*

Thus no sign appears: $(*)$ never compares $\hat{\beta}$ with the Hodge dual. The chain of [4, Corollary 8.5], normalised by $\beta_{n+1} = 1$ and $\alpha_k = \hat{\beta}_k$, is itself unique: the type of normalised chains is contractible, by the same uniqueness argument in coordinates (a factor γ in $\bigwedge^q(B) = \gamma \hat{\beta}^T$ is unique when $\hat{\beta}$ is regular).

Definition 7. $\mathrm{Det}(F) := R_1$, a line, and $a_{\mathrm{Det}} := a_1 : \mathrm{Det}(F) \to \bigwedge^{r_1} F_0$, for the unique structure.

Theorem 8. *Under the hypotheses of Theorem 5, assume moreover that F_0 has rank r_1. Then $\bigwedge^{r_1} F_0$ is a line, a_{Det} is injective, and for any bases, a_{Det} at the basis element of $\mathrm{Det}(F)$ determined by the bases is $\delta\,\omega$, where ω is the basis element of $\bigwedge^{r_1} F_0$ and δ is the Cayley determinant of* [4]*. The image of a_{Det}, i.e. the ideal of values $\lambda(a_{\mathrm{Det}}(\varphi))$ for λ a linear form, is the ideal (δ); it is the gcd ideal of $D_{r_1}(f_1)$: the elements divisible by some greatest common divisor.*

Proof. In coordinates the structure is the one built from the chain of [4, Corollary 8.5], by uniqueness; its last map sends the basis element to $\delta\omega$. Every φ is c times the basis element, so the image is (δ), and δ is a regular greatest common divisor of the r_1-minors [4, Theorem 8.7]. Injectivity: $\mathrm{top}(a_{\mathrm{Det}}(c \cdot \mathrm{basis})) = c\delta$ and δ is regular. The gcd ideal is defined without choosing a gcd, and equals (g) for every gcd g. □

5. Resolutions of an ideal and the greatest common divisor

A *finite free resolution* of the ideal $(a_1, \dots, a_N)$ is an exact complex of finite free modules

$$0 \to F_n \xrightarrow{f_n} \cdots \xrightarrow{f_2} F_1 \xrightarrow{f_1} F_0$$

(f_n injective, exact at each F_k, $1 \le k < n$) with a linear bijection $\iota : F_0 \cong R$, such that the image of ιf_1 is the ideal $(a_1, \dots, a_N)$. The terms are only merely free: no basis is part of the data.

Proposition 9. *Assume $1 \neq 0$. The rank data of a resolution, namely the ranks s_k of the F_k and numbers r_k with $s_k = r_k + r_{k+1}$ such that f_k is stable of rank r_k, form a proposition, and are merely given, hence given. The hypotheses of Theorem 5 hold. Moreover, $r_1 = 1$ unless all a_j are 0.*

Proof. Uniqueness: free ranks and ranks of stable maps are unique since $1 \neq 0$. Existence: after choosing bases, [4, Theorem 6.8] gives the ranks, stability and $\mathrm{Gr}(D_{r_k}(f_k)) \ge k$; by Proposition 2 these hold without bases. Since $s_0 = 1$, $r_1 \in \{0, 1\}$ or $1 = 0$: if $r_1 \ge 2$, $D_{r_1}(f_1)$ is regular but $\bigwedge^2$ of a rank-one module is 0. If $r_1 = 0$ then $\bigwedge^1 f_1 = 0$, so every 1×1 minor of f_1 is 0 in any bases, hence $\iota f_1 = 0$ and all a_j are 0. The case split is on the number r_1, so no choice is involved. □

Let now (a_j) be a nonzero ideal: not all a_j are 0. Then $r_1 = 1$ and Theorem 8 applies. The bijection ι, followed by $R \cong R^1$, is a basis of F_0 that comes with the resolution, and the top coordinate in that basis is an isomorphism $J : \bigwedge^1 F_0 \cong R$. Define

$$a_1 := J \circ a_{\mathrm{Det}} : \mathrm{Det}(F) \to R.$$

This is the map $a_1 : R_1 \to \bigwedge^1 F_0 = R$ of [2, Theorem 3.1].

A *greatest common divisor* of (a_j) is g dividing all a_j such that every common divisor of the a_j merely divides g; it is *regular* if $gx = 0$ implies $x = 0$. For regular g the cofactors are unique, so "g is a regular gcd" is a proposition; write RegGcd for the set of regular gcds.

Theorem 10. *Let F be a finite free resolution of a nonzero ideal $(a_1, \dots, a_N)$, $1 \neq 0$. Then $a_1 : \mathrm{Det}(F) \to R$ is injective, its image is the gcd ideal of (a_j), and*

$$\mathrm{Bases}(\mathrm{Det}F) \;\simeq\; \mathrm{RegGcd}, \qquad \varphi \mapsto a_1(\varphi).$$

Both sides are $R^\times$-torsors and the map is equivariant.

Proof. J is a bijection, so a_1 is injective by Theorem 8, and its image is the image of a_{Det}, since every linear form on the line $\bigwedge^1 F_0$ is J times a scalar. The image of a_{Det} is the gcd ideal of the 1×1 minors of f_1 in a basis, and with the identification ι these generate (a_j), so the gcd ideal is that of (a_j) (a gcd depends only on the ideal).

For any injective linear $a : K \to R$, $\varphi \mapsto a(\varphi)$ is a bijection between the bases of K and the regular generators of the image of a. If φ is a basis, $a(\varphi)x = a(x\varphi) = 0$ gives $x\varphi = 0$, so $x = 0$; and $a(c\varphi) = a(\varphi)c$. Conversely a regular generator g is $a(\varphi)$ for a unique φ (a injective, unique choice); $c\varphi = c'\varphi$ gives $gc = gc'$, so $c = c'$; and $a(\psi) = gt = a(t\varphi)$ gives $\psi = t\varphi$.

It remains that the regular generators of the gcd ideal are the regular gcds. A gcd g generates it. Conversely let g be a regular generator and g_0 a gcd (merely, by Theorem 8). Then $g_0 = gt$ and $g = g_0 u$; so g divides the a_j, and every common divisor divides g_0, hence g. Both sides are propositions in g, so the two subtypes of R are equal. $R^\times$ acts freely and transitively on regular gcds ($g' = gu$, $g = g'v$ give $vu = 1$ since g is regular), and $a_1(\varphi u) = a_1(\varphi)u$. □

Constructively one cannot split into "all a_j are 0" and "not all a_j are 0". A resolution supplies the case distinction: the number $r_1 \in \{0, 1\}$ of Proposition 9 is computed from the resolution.

Proposition 11. *Let F be a finite free resolution of $(a_1, \dots, a_N)$, $1 \neq 0$. Then either all a_j are 0 (when $r_1 = 0$), or the regular greatest common divisors of the a_j form an $R^\times$-torsor, namely the values of a_1 at the bases of $\mathrm{Det}(F)$ (when $r_1 = 1$).*

Corollary 12. *If $1 \neq 0$ and every ideal (a, b) has a finite free resolution, then for all a, b: either $a = b = 0$, or the regular gcds of a, b form an $R^\times$-torsor. In both cases a, b merely have a gcd, and the gcd ideal is merely principal.*

This is [4, Corollary 8.10] with the gcd as a canonical object. For the zero ideal the gcd is 0, which is not regular unless $1 = 0$, so Theorem 8.9 of [4] is to be read with this case distinction.

6. Independence of the resolution

Lemma 13. *Let $a : K \to R$ and $a' : K' \to R$ be injective linear maps with the same image. There is a unique linear isomorphism $e : K \cong K'$ with $a'e = a$.*

Proof. For $x \in K$, $a(x)$ is in the image of a', so merely $a(x) = a'(y)$; such a y is unique since a' is injective, so it is given by unique choice, and we put $e(x) = y$. Linearity: $a'(e(x + x')) = a(x) + a(x') = a'(e(x) + e(x'))$, and a' is injective; similarly for scalars. The inverse is built in the same way from a' and a. If also $a'e' = a$, then $a'(e'(x)) = a'(e(x))$, so $e' = e$. □

Theorem 14. *Let $1 \neq 0$ and let F, G be finite free resolutions, of any lengths, of the same nonzero ideal $(a_1, \dots, a_N)$. There is a unique isomorphism $e_{F,G} : \mathrm{Det}(F) \cong \mathrm{Det}(G)$ with $a_G \circ e_{F,G} = a_F$, where a_F, a_G are the maps a_1 of F and G. Moreover $e_{F,F} = \mathrm{id}$ and $e_{G,H} \circ e_{F,G} = e_{F,H}$.*

Proof. By Theorem 10, a_F and a_G are injective with image the gcd ideal of (a_j), which depends only on the a_j. Lemma 13 gives $e_{F,G}$. The identity and $e_{G,H}e_{F,G}$ satisfy the defining equations of $e_{F,F}$ and $e_{F,H}$, so they are equal to them by uniqueness. □

So the line with its injection depends canonically on the ideal, not on the resolution.

Fix N, a and the length n. An *isomorphism* between two resolutions (F, f, ι) and (F', f', ι') of (a_j) is a family of linear bijections $\varphi_k : F_k \cong F'_k$ with $\varphi_{k-1} f_k = f'_k \varphi_k$ and $\iota' \varphi_0 = \iota$.

Theorem 15. *An isomorphism of resolutions $F \cong F'$ is an equality $F = F'$. Consequently, for a nonzero ideal, transport along it gives $e : \mathrm{Det}(F) \cong \mathrm{Det}(F')$ with $a'_1 \circ e = a_1$, and $e = e_{F,F'}$.*

Proof. By induction on isomorphisms (the type of families $\sum_{F'} \prod_k (F_k \cong F'_k)$ is contractible) we may assume $F'_k = F_k$ and $\varphi_k = \mathrm{id}$. Then $f'_k = f_k$ and $\iota' = \iota$ pointwise, hence as linear maps, and the remaining data (injectivity, exactness, the image condition, the mere existence of bases) are propositions. For an equality $F = F'$ the second statement holds by transport from $e = \mathrm{id}$, and $e = e_{F,F'}$ by the uniqueness in Theorem 14. □

Every construction on resolutions is natural in the same way. The same argument for rings gives, for instance, the invariance of the true grade under ring isomorphisms as a single transport.

7. Wiebe's theorem without bases

Let $V : F \to F$ be an endomorphism of a finite free module of rank n. $\bigwedge^n V$ acts on the line $\bigwedge^n F$; the scalar Δ with $\bigwedge^n V(u) = \Delta u$ for all u is unique (the line has a basis u_0, and $\Delta = J(\Delta u_0)$ for an isomorphism J with $J(u_0) = 1$), and exists: in a basis with matrix A, $\bigwedge(A)(\omega) = \det(A)\,\omega$. So $\Delta = \det V$ is defined without bases and is $\det A$ in every basis.

For linear forms $\alpha, \beta : F \to R$ and a finitely generated ideal J_0, write (J_0, β) for the ideal of x merely congruent to some $\beta(v)$ modulo J_0, and similarly (J_0, β, α) and (J_0, β, Δ).

Theorem 16. *Assume $\beta \equiv \alpha \circ V$ modulo J_0 and $\mathrm{Gr}(\beta) \geq n$ modulo J_0 (for the coefficients of β in one, equivalently every, basis). Then modulo (J_0, β): if $x\Delta \in (J_0, \beta)$ then $x \in (J_0, \beta, \alpha)$, and if $x\alpha(w) \in (J_0, \beta)$ for all w then $x \in (J_0, \beta, \Delta)$.*

Proof. The conclusions are propositions; choose a basis. A linear form λ is $v \mapsto \sum_k c_k (ev)_k$ with $c_k = \lambda(e^{-1}\delta_k)$, so the image of λ is the ideal (c). With a, b the coefficient vectors of α, β and A the matrix of eVe^{-1} (column j holds the coordinates of the image of e_j), $b_j = \beta(e^{-1}e_j) \equiv \alpha(Ve^{-1}e_j) = \sum_i a_i A_{ij}$ modulo J_0, and $\Delta = \det A$. This is the situation of [5, Theorem 11] relative to J_0, with $V = A$, which gives the conclusions for the lists (J_0, b), (J_0, b, a), (J_0, b, Δ); these are the ideals of the forms. □

The statement involves no basis, so it is invariant under any change of basis; in [5] the corresponding invariance under elementary operations is part of the proof, which is used unchanged here.

8. What is proved and what is checked

All statements above are checked in the proof assistant Unit (`https://github.com/coquand/unit`), designed by T. Coquand and implemented by Claude (Anthropic), in the directory `sag/cffru`. The matrix-level results of [4] and [5] are formalised in `sag/cffr`, `sag/koszul` and `sag/wiebe`, and the structure identity principle in `sag/sip`; they are used unchanged. The whole tree checks in a few seconds.

statement	file
structure identity (generic; modules, rings)	`sag/sip`: `SipGeneric`, `ModuleSIP2`; `RingSIP`
$\bigwedge M$, $\bigwedge^k M$, the top line, Hom of lines	`ExteriorFree`, `TopLine`, `Lines`
Proposition 2, ranks	`DkFree`, `Ranks`
Proposition 6	`Dictionary`
Theorem 5	`StageStar`, `Structure`
Theorem 8	`DetLine`
Proposition 9	`DetFFR`
Theorem 10, torsors	`GcdTorsor`, `DetGcd`
Proposition 11, Lemma 13, Theorem 14	`DetCompare`
Theorem 15	`NaturalFFR`, `DetCompare`
contractible chains of [4, Cor. 8.5]	`Factorisations`
det of an endomorphism, Theorem 16	`DetEnd`, `WiebeFree`

Not treated: part (b) of [2, Theorem 3.1] ($\operatorname{Rad} I(a_k) = \operatorname{Rad} I(f_k)$), whose proof uses prime ideals; and a proof of Wiebe's theorem that does not pass through coordinates.

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